iTaskSense: Task-Oriented Object Detection in Resource-Constrained Environments

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***Abstract*—Task-oriented object detection is increasingly essential for intelligent sensing applications, enabling AI systems to operate autonomously in complex, real-world environments such as autonomous driving, healthcare, and industrial automation. Conventional models often struggle with generalization, requiring vast datasets to accurately detect objects within diverse contexts. In this work, we introduce iTask, a task-oriented object detection framework that leverages large language models (LLMs) to generalize efficiently from limited samples by generating an abstract knowledge graph. This graph encapsulates essential task attributes, allowing iTask to identify objects based on high-level characteristics rather than extensive data, making it possible to adapt to complex mission requirements with minimal samples.**

iTask addresses the challenges of high computational cost and resource limitations in vision-language models by offering two configuration models: a distilled, task-specific vision transformer optimized for high accuracy in defined tasks, and a quantized version of the model for broader applicability across multiple tasks. Additionally, we designed a hardware acceleration circuit to support real-time processing, essential for edge devices that require low latency and efficient task execution. Our evaluations show that the task-specific configuration achieves a 15% higher accuracy over the quantized configuration in specific scenarios, while the quantized model provides robust multi-task performance. The hardware-accelerated iTask system achieves a 3.5x speedup and a 40% reduction in energy consumption compared to GPU-based implementations. These results demonstrate that iTask 's dual-configuration approach and situational adaptability offer a scalable solution for task-specific object detection, providing robust and efficient performance in resource-constrained environments.

I. Introduction

Task-oriented object detection [1], [2] is crucial for a wide range of applications, from autonomous driving and surveillance to healthcare and industrial automation [2], [3]. Unlike general object detection, task-oriented detection focuses on identifying and localizing objects within a specific contextual frame, allowing AI systems to make informed decisions and initiate context-driven actions [4]. Meeting the demands of task-oriented detection often requires models to adapt to complex mission requirements, which are frequently defined by abstract characteristics rather than explicit labels. This dependency on nuanced contextual understanding challenges traditional models, which typically rely on large, annotated datasets to achieve high accuracy [5]. Furthermore, implementing these models in real-world scenarios introduces additional constraints, such as computational and memory efficiency, particularly in resource-constrained environments like edge devices.

Another challenge is that traditional vision-language models, such as MDETR [6], used to solve tasks [7] are very resource-intensive, consuming high amounts of memory and power. This is particularly evident in intelligent sensing applications, where edge devices must operate efficiently with limited computational resources and potentially minimal data samples [8]. To address these demands, we present *intelligent Task Oriented Aritficial Intelligence (*iTask*)*, a task-oriented object detection framework designed to excel in complex missions and adapt effectively to a wide range of real-world tasks.

Consider a complex mission scenario where a system must identify any object that could be used to open a bottle of beer. In a traditional

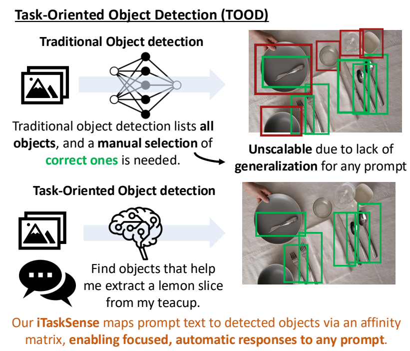

Fig. 1: **iTask framework for task-oriented object detection**: Unlike traditional detection, iTask uses an affinity matrix for prompt-based, context-aware responses, enhancing precision and scalability across domains while tackling high computational demands and complex object semantics.

machine learning context, a model would need vast amounts of labeled data to capture all potential objects capable of performing this action, such as bottle openers, lighters, or even certain types of spoons [2]. However, real-world scenarios often require generalizing from very few samples, which is nearly impossible for traditional methods without comprehensive datasets. iTask addresses this challenge by using a large language model (LLM) to generate an abstract knowledge graph that encapsulates the essential attributes required to perform the task, such as "firm leverage" or "rigid edge". This high-level understanding allows iTask to define these characteristics and identify objects that meet them, enabling generalization even with minimal data samples.

For example, in a scenario where the goal is to extract a lemon slice from a glass, iTask can analyze objects that share attributes relevant to the task, as shown in Figure 1. While a knife, fork, or spoon might be typical choices, an unexpected tool like a pen could also serve the purpose, yet it may not appear in conventional datasets unless trained with vast samples. By leveraging the LLM-generated knowledge graph, iTask can identify and rank objects based on their feature alignment with task requirements, selecting the most suitable option even in cases where there is no perfect match or multiple candidates.

This capability is further enhanced through iTask's situational awareness, which enables real-time mission adaptability. When deployed on edge devices, iTask can interact dynamically with its environment to analyze content based on specific mission objectives. For instance, if an edge device is instructed to "identify objects capable of providing leverage," iTask can assess available objects and select those that match the task-defined attributes . This real-time adaptability not only reduces dependency on extensive datasets but also significantly improves resource efficiency by focusing on task-relevant data, optimizing computational load, and conserving energy.

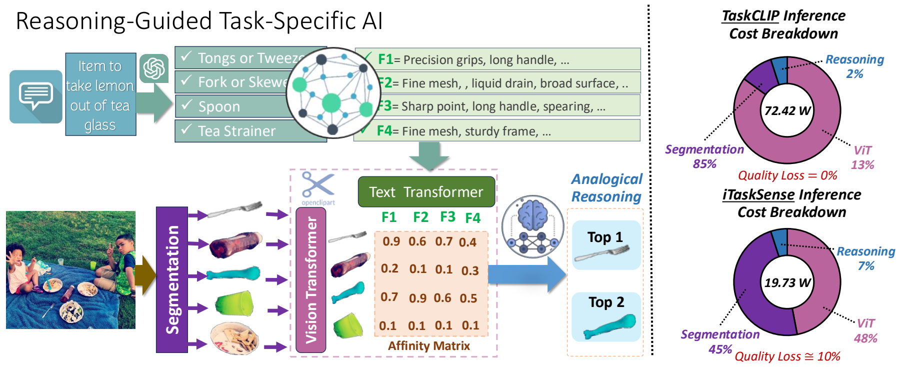

Fig. 2: **The iTask framework for reasoning-guided task-specific object selection and efficient hardware acceleration:** Given the prompt, segmented objects are encoded and matched to functional features (F1–F4) via an affinity matrix. Analogical reasoning ranks items (e.g., forks, tweezers) based on task-relevant attributes. **Hardware optimization for taskCLIP inference:** Smaller CLIP models reduce cost but may lose quality; knowledge distillation improves their accuracy. A unified ASIC design accelerates segmentation (convolution-based) and ViT (attention-based) tasks on the same hardware, achieving significant computational efficiency.

To overcome the limitations of traditional vision-language models in intelligent sensing, iTask is built with two configurations tailored for different mission requirements. The first configuration uses knowledge distillation to create a smaller, task-specific vision transformer that maximizes detection performance for specific tasks. While smaller models reduce computational costs, they often lose quality compared to larger models. iTask addresses this challenge by employing knowledge distillation techniques to make smaller CLIP models as accurate as their larger counterparts, ensuring high-quality performance without the associated cost overhead. The second configuration employs quantization to compress the model, enabling versatility across multiple tasks with reduced computational demands [9], [10]. These configurations allow iTask to provide efficient task-oriented detection in a wide variety of real-world scenarios while maintaining adaptability and responsiveness.

Additionally, a hardware circuit was designed to accelerate iTask, enabling rapid processing near the sensor and meeting the real-time demands of intelligent sensing environments. Since segmentation and vision transformers (ViT) represent fundamentally different network architectures—segmentation being convolution-based and ViT attention-based—specific hardware accelerators are required for each. To address this challenge, we propose a unified ASIC architecture that accelerates both ViT and segmentation tasks on the same hardware platform, significantly improving computational efficiency for taskCLIP inference. This design is essential for edge-based AI applications, where high-speed processing and minimal latency are critical. Through this combined approach of model optimization, situational adaptability, and hardware acceleration, iTask delivers a scalable, efficient solution tailored for resource-constrained environments where robust task-specific detection is essential.

The contributions of this paper are threefold: (1) the introduction of iTask, a framework that integrates LLM-generated knowledge graphs for task-specific generalization with minimal samples, achieving a 15% improvement in accuracy over traditional quantized configurations in specific scenarios; (2) dual configuration models, including knowledge distillation and quantization, for efficient task-specific and multi-task object detection; and (3) a hardware acceleration circuit that enables real-time, low-latency performance in intelligent sensing applications, delivering a 3.5x speedup and a 40% reduction in energy consumption compared to standard GPU-based implementations These advancements make iTask a scalable, efficient solution for complex, task-oriented detection in resource-constrained environments.

II. Related Work

A. Vision-Language Models

Vision-Language Models (VLMs) integrate visual and textual data for multimodal representation, bridging visual content with textual queries. VinVL, for example, achieves high performance in multimodal tasks, utilizing a vision-language-enhanced object detection model [11]. MiniVLM adapts VLM principles for resource-constrained environments using a Two-stage Efficient Extractor (TEE) and a compact transformer-based fusion module inspired by EfficientDet and MiniLM [12]–[14]. Despite their strengths, VLMs remain on the critical path for decision-making, where costly and non-interpretable outputs present challenges. Guided by MiniVLM's efficiency, we design iTask to combine intelligent sensing with LLMs, addressing task-oriented detection challenges in edge computing scenarios while maintaining interpretability and computational efficiency.

B. Task-Oriented Object Detection

Task-oriented object detection [1], [2] focuses on identifying objects relevant to specific tasks, requiring robust reasoning to address complex demands. Task-aligned One-stage Object Detection (TOOD) enhances classification and localization by generating task-interactive features to reduce spatial misalignment [15]. Similarly, Wang et al. [16] developed a task-oriented degraded image enhancement network, achieving precision improvements of up to 1.6% on degraded datasets like URPC2020 [17]. However, achieving efficiency in TOOD remains challenging due to the reliance on large, compute-intensive models. To address these issues, our work integrates TOOD with intelligent sensing and lightweight LLMs to enhance detection precision, efficiency, and generalization in resource-constrained edge environments, while leveraging hardware optimizations for task-specific inference.

C. Intelligent Sensors

Intelligent sensors have become a transformative innovation in autonomous systems by enabling near-sensor data processing, which reduces latency, enhances privacy, and minimizes bandwidth through selective data acquisition and compression. Huang et al. [18] demonstrated notable energy savings using near-sensor AI models that process only frames of interest. Similarly, Zhang et al. [19] showed that adaptive thresholds improve compression and reconstruction accuracy, while Lu et al. [20] achieved dynamic threshold adjustments, enhancing performance. Despite these advancements, intelligent sensing faces significant challenges in algorithm-hardware co-optimization [21]. Our framework addresses these challenges by identifying regions of interest (ROIs) that enable lightweight sensors to filter irrelevant data efficiently. Combined with task-oriented object detection and VLMs, this approach significantly improves the energy efficiency and adaptability of intelligent sensing systems. To achieve real-time detection and reasoning in edge environments with limited resources, we design and implement a domain-specific ASIC accelerator to enhance our model's performance.

D. Hardware Design for Intelligent Sensing

Hardware design is critical for intelligent sensing applications, where energy efficiency and accuracy are constrained by sensor type, data resolution, and model complexity. Taha et al. [22] developed a low-power wireless transmission architecture that activates only critical sensors, leveraging near-sensor deep learning to balance efficiency and accuracy. Similarly, Ballard et al. [23] optimized spectral encoding for hyperspectral satellites, significantly reducing costly data transmission overheads. Huang et al. [18] designed a near-sensor object detection model capable of capturing frames of interest, eliminating the need for high-bandwidth transmission. However, hardware limitations often prevent running multiple algorithms like segmentation and ViT on the same platform. To overcome this, our work introduces a unified ASIC design that supports both convolution-based segmentation and attention-based ViT tasks. This architecture enables efficient taskCLIP inference by combining task-oriented detection with energy-efficient hardware acceleration, ensuring scalability and adaptability in dynamic edge environments.

III. Framework

A. Core Design of iTask for Efficient Task-Driven Detection

Ignited by prior research [1], iTask initially integrates the open-vocabulary capability of OpenAI CLIP with the reasoning power of LLM. Subsequently, it emphasizes model compression along with hardware and software co-design for deploying the framework on edge. It employs a multi-stage process to bridge the semantic gap between task affordances and visual object features, optimizing both accuracy and efficiency for edge computing scenarios.

Figure 2 presents the framework of iTask. The first stage of the iTask framework involves general object detection using pre-trained models to identify potential object candidates. In this step, bounding boxes and object features are generated for all items in the scene. To ensure effective reasoning and selection, iTask integrates an LLM-guided parsing mechanism to extract task-specific affordance attributes, such as "precision grip" or "long handle," relevant to the given prompt. These attributes are encoded as visual features, enabling a unified embedding space for images and text.

In the second stage, task-specific object selection is performed using a transformer-based aligner module. Inspired by TaskCLIP, the aligner recalibrates embeddings from the vision-language model (VLM), addressing misalignment between object images (e.g., nouns) and their visual attributes (e.g., adjective phrases). The aligner's multi-layer design incorporates self-attention and cross-attention mechanisms to refine the embeddings, ensuring they accurately represent task-specific affordances. This recalibration improves the semantic alignment and ensures high-quality reasoning-driven selection of objects.

Finally, iTask employs a trainable scoring function to process the affinity matrix generated by embedding alignment. This scoring function ranks objects based on their suitability for the specified task,

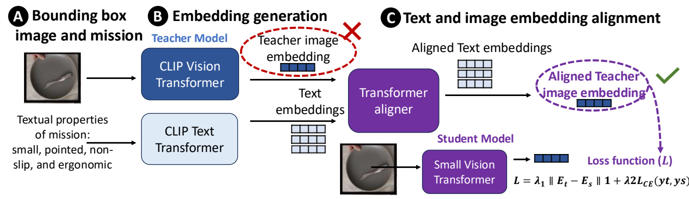

Fig. 3: Diagram of the iTask framework with knowledge distillation. The teacher model (CLIP ViT) generates image and text embeddings distilled to a smaller ViT model. A transformer aligner enhances alignment for task-specific object detection, addressing low performance when using pre-aligned image embeddings.

enabling robust task-oriented detection. By leveraging the inherent capabilities of VLMs and combining them with reasoning mechanisms, iTask achieves scalable and efficient detection, outperforming single-stage approaches in terms of generalizability and adaptability.

B. Lightweight Adaptation of TaskCLIP

To address the computational demands of the iTask framework, we introduce a strategy for lightweight adaptation, optimizing the performance of large vision-language models (VLMs) like TaskCLIP for resource-constrained environments. This approach employs two key techniques, affinity mimicking and weight inheritance, to compress and accelerate the model while maintaining robust task-oriented detection capabilities. Affinity mimicking allows the lightweight model to replicate the cross-modal feature alignment of a larger VLM. The image-text affinity scores generated by the large model, A_{large}, are distilled into the lightweight model, A_{light}, using the following loss function:

$$\mathcal{L}_{\text{affinity}} = \|A_{\text{large}} - A_{\text{light}}\|_F^2,$$

where $\|\cdot\|_F$ denotes the Frobenius norm. This ensures that the lightweight model retains essential semantic understanding for task-specific detection. To improve initialization, critical weights from the large model, θ_{large}, are transferred to the lightweight model, θ_{light}, using a selective masking strategy:

$$\theta_{\text{light}} = M \odot \theta_{\text{large}},$$

where M is a learnable mask identifying important weights, and $\odot$ denotes element-wise multiplication. By integrating these techniques, we significantly reduce the model size and computational complexity of iTask while preserving its task-specific detection performance. As shown in the inference cost breakdown on the right side of Figure 2, the proposed approach reduces reasoning and segmentation costs, enabling efficient and accurate task-driven detection across diverse real-world applications.

While smaller CLIP models reduce computational cost, they may suffer from quality degradation; however, knowledge distillation effectively enhances their accuracy. To further support this efficiency, a unified ASIC design accelerates both segmentation (convolution-based) and ViT (attention-based) tasks on the same hardware. This hardware optimization achieves significant computational efficiency by seamlessly integrating the distinct requirements of these tasks, making TaskCLIP inference highly suitable for resource-constrained environments such as edge computing systems.

C. Knowledge Distillation for Task-Specific Precision

To address the computational demands of the iTask framework while maintaining high task-specific performance, we propose a knowledge distillation technique that uses EfficientNet as a starting point to train a lightweight student model as a substitute for the vision transformer in CLIP, as shown in Figure 3. This approach targets the challenge of achieving the quality of large CLIP models without

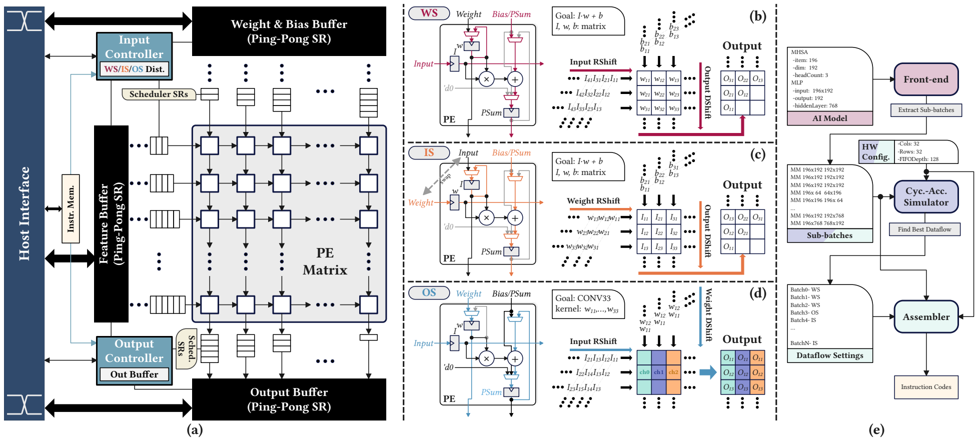

Fig. 4: The hardware/software stack for iTask acceleration. (**a**) The proposed systolic-array-based hardware accelerator architecture. (**b**) Weight stationary dataflow implemented with our PE design. (**c**) Input stationary dataflow implemented with our PE design. (**d**) Output stationary dataflow implemented with our PE design. (**e**) The compiler stack for the proposed accelerator architecture.

incurring their computational costs. Our distilled model balances accuracy and efficiency effectively, making it suitable for resource-constrained environments.

In this setup, the original CLIP vision transformer serves as the teacher model, guiding the training of a compact student model. The student model employs EfficientNet's compound scaling methodology [24], which systematically balances network depth, width, and resolution to achieve both efficiency and accuracy. This ensures that the student captures essential visual patterns while aligning semantically with textual embeddings. The distillation process minimizes a combined loss function:

$$\mathcal{L} = \lambda_1 \|E_t - E_s\|_1 + \lambda_2 \mathcal{L}_{\text{CE}},$$

where E_t and E_s are the embeddings from the teacher and student models, respectively, and $\mathcal{L}_{\text{CE}}$ is the cross-entropy loss between the output distributions of the two models. The L1 loss preserves the semantic alignment between teacher and student embeddings, while the cross-entropy loss ensures consistency in task-specific predictions. To stabilize training and reduce representational gaps, knowledge is transferred progressively in multiple stages. This approach ensures that the student model iteratively refines its representations while avoiding overfitting the teacher's embeddings. As depicted in Figure 3, a task-specific transformer aligner further refines the student model's embeddings to enhance task-related affordances. This recalibration addresses subtle visual details crucial for precise object detection in resource-constrained environments.

IV. Hardware Acceleration

Accelerating iTask benefits when utilizing unified hardware architecture accelerating both CNN and ViT. This requires a balance between general-purpose and domain-specific nuances. The existing parallel processing hardware, such as GPU and FPGA, have negatively biased nuances in terms of iTask acceleration. The former is biased toward general-purpose nuance, enabling scalability but lacking energy efficiency. The latter can organize the dataflow more flexibly but lacks computing power due to its intrinsic architectural limitations, thus, optimized to to accelerate specific architectures. Moreover, current domain-specific accelerators, whether FPGA-based or ASIC-based, typically concentrate solely on either CNN or VIT acceleration, rendering them ill-suited to support state-of-the-art vision language models. To solve all these challenges, we designed the hardware accelerator for ASIC implementation, expanding the scalability to diverse task-oriented applications for iTask.

Fig. 4 presents the hardware/software stack designed to accelerate the iTask. The stack consists of the systolic array-based hardware accelerator, and the domain-specific compiler to generate the instruction codes executed on the accelerator.

A. Hardware Accelerator

For most computation kernels in ViT and CNN workloads, we designed the systolic array architecture with 8-bit MAC in each PE, which enhances data reusability and overcomes memory-bound limitations. The unified PE design in Fig. 4(b)-(d) enables to support three dataflows, including the weight stationary (WS), input stationary (IS), and output stationary (OS), with proper organization of multiplexers (MUXes) in weight and partial sum (PSum) registers. This ensures the optimal dataflow execution for each computation layer, practically accelerating diverse ViT and CNN configurations.

Our architecture organized the input buffers (feature, weight, bias) as shift registers (SRs) with a ping-pong strategy, enabling seamless feeding to the systolic array (see Fig. 4(a) PE Matrix). This avoids using SRAM macros in ASIC design, as SRAM would present the critical-path delay due to its addressing circuit, making it a bottleneck in our design, which adopts 8-bit operations to ensure high-speed operations. This not only increases the maximum frequency but also the energy efficiency, as the ping-pong SRs do not require much memory compared to the SRAM buffer. Another notable characteristic is the adoption of scheduler SRs (see Fig. 4(a)) in both I/O buffers. The scheduler SRs automatically generate the sequence required for the systolic array computation, which is presented in Fig 4(b)-(d) for each dataflows, to implement ViT and CNN operations from the original matrix data structure. This eliminates the host computer's data re-ordering workloads, including pre- and post-processing.

B. Compiler Stack

The compiler stack, illustrated in Fig. 4(e), organizes a three-stage operation. The front-end part extracts the lower-level computation layers, e.g., matrix multiplication or convolution, from the high-level machine learning model architectures, including CNN and ViT. This enables the swift diversification of the model's hyperparameters.

The computation layers extracted from the front-end are cast to the cycle-accurate simulator with systolic array configuration (M×N). The simulator executes each computation layer with supporting dataflows,

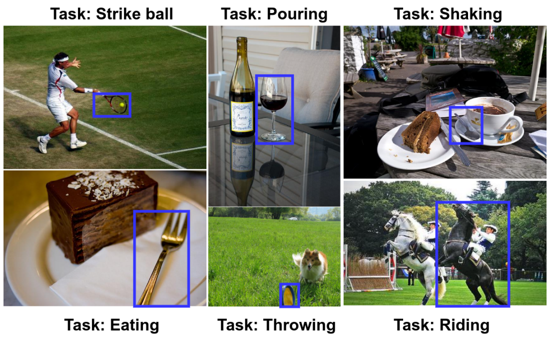

Fig. 5: Examples of task-oriented object detection with iTask, where specific objects are identified based on the task context. Tasks include 'Strike ball,' 'Pouring,' 'Shaking,' 'Eating,' 'Throwing,' and 'Riding,' with detected objects highlighted in blue. This approach enables context-aware object recognition tailored to task-specific requirements.

deriving the optimal dataflow for each layer. This simulator is developed upon Scalesim [25], modifying the cycle calculation to apply our accelerator's data-path using ping-pong SR FIFO.

Finally, the assembler produces the instruction code binary to execute given machine learning tasks using specific hardware configurations. With the optimized dataflow for each layer from the simulator executed using the same configurations, the optimal code for the ML workload is generated.

V. Evaluation

This section presents the evaluation of the iTask framework across various task-oriented object detection tasks, highlighting the performance of our original configuration and quantized and knowledge distillation configurations. We use AP@0.5 as our primary metric for evaluating task-oriented object detection, as it offers a suitable balance between precision and recall, especially relevant for task-specific applications that prioritize accurate identification over general object detection. AP@0.5 means at a 50% Intersection over the Union (IoU) threshold, is chosen as the primary evaluation metric due to its robustness in balancing precision and recall for task-specific object detection. The machine learning side experiments were conducted on an NVIDIA RTX 4090, with hardware synthesis performed through $28nm$ CMOS technology using Synopsys Design Compiler, specifically 0.81v @ 125C corner. The accelerator is designed with Chisel [26] for swift deployment across diverse systolic array configurations. In this evaluation, we selected the 128 $\times$ 128, a widely used configuration in other works. Table III presents the configurations and results. We also analyze the energy and speed our hardware accelerator achieves in task-oriented scenarios. In task-oriented applications, the focus is often on detecting objects that meet specific criteria rather than maximizing accuracy across all detected instances. AP@0.5 allows us to prioritize objects that align closely with each task's intended functionality or affordances, making it ideal for scenarios where the utility of an object (e.g., can it hold flowers?) is more critical than precise localization. Higher IoU thresholds, while beneficial for general detection, may penalize task-oriented models by undervaluing objects that partially meet task requirements, whereas AP@0.5 provides a balanced metric for functional correctness in such cases.

A. iTask *Configuration Performance*

Table I showcases the performance of state-of-the-art (SOA) models for task-oriented object detection on COCO tasks, including our iTask implementation and its variations. Among these, iTask* (YOLO) achieves the highest average performance (45.01), demonstrating superior task-specific accuracy while maintaining competitive results across all tasks. This establishes our iTask framework as a leader in the SOA for task-oriented object detection.

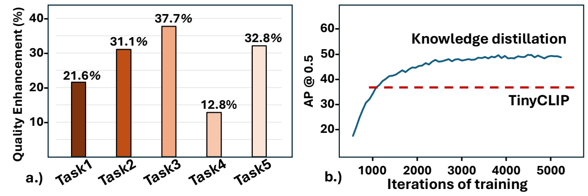

Fig. 6: (a) The graph shows the quality improvement of the student model implemented using knowledge distillation compared to tiny-CLIP, (b) convergence of TinyCLIP and Knowledge distillation at Task 1.

Our iTask implementation includes two configurations: a region-based RCNN approach and an efficient YOLO-based single-shot detector. iTask (YOLO) achieves an average performance of 44.51, while iTask* (YOLO) further improves accuracy with task-specific tuning. These results highlight the robustness and adaptability of iTask across diverse detection scenarios.

With iTask setting the SOA, the next step focuses on quantized versions of the model to reduce computational cost, memory usage, and inference time. The goal is to create lightweight, fast, and accurate configurations suitable for deployment in resource-constrained environments such as edge devices and real-time applications, extending the impact of iTask to practical use cases.

The lightweight version, iTask Tiny, introduces a trade-off between accuracy and computational efficiency. Across tasks, iTask Tiny configurations exhibit a quality loss ranging from approximately 8% to 12% relative to iTask (YOLO), depending on the specific quantization settings. For example, more aggressive quantization configurations like (8-4) experience higher performance degradation, while less aggressive configurations such as (10-16) achieve reduced quality loss. Despite this degradation, iTask Tiny retains sufficient accuracy for many task-specific applications, providing a highly efficient alternative for deployment in resource-constrained environments.

This quality loss reflects the inherent challenges in balancing lightweight design and task-specific accuracy. However, iTask Tiny demonstrates that, through careful quantization and optimization, lightweight models can still deliver effective task-oriented object detection while significantly reducing computational requirements. These results highlight the potential for further refinement of quantized models to bridge the gap between efficiency and performance for real-world systems.

B. *Knowledge Distillation Results*

Figure 5 illustrates examples of task-oriented object detection using iTask for the lightweight version of our model. Specific objects are identified based on the context of various tasks, such as *'Strike ball,' 'Pouring,' 'Shaking,' 'Eating,' 'Throwing,'* and *'Riding.'* Detected objects are highlighted in blue, demonstrating the model's ability to perform context-aware object recognition tailored to task-specific requirements. This contextual adaptability underscores the effectiveness of the lightweight iTask framework in aligning object detection with task-specific affordances.

Moving to the quantitative analysis, Figure 6 provides deeper insights into the performance improvements achieved through knowledge distillation. Subfigure (a) compares the quality enhancement of EfficientViT, implemented using knowledge distillation, against TinyCLIP. The results clearly show that knowledge distillation leads to a significantly higher ceiling in terms of performance, with EfficientViT outperforming TinyCLIP in achieving task-specific quality

TABLE I: Comparison of different models across tasks with iTask variations.

Task	**GGNN** [27]	**TOIST** [28]	**TOIST*** [28]	**iTask (RCNN)** [1]	**iTask (YOLO)** [1]	**iTask* (YOLO)** [1]
1	36.6	44.0	45.8	44.23	44.23	48.31
2	29.8	39.5	40.0	43.71	43.71	46.23
3	17.2	23.5	26.9	31.65	31.64	30.71
4	43.6	52.8	58.3	65.54	65.54	62.02
5	21.0	23.0	32.5	37.46	37.46	37.74
Mean	29.64	36.56	40.7	44.52	44.51	45.01

TABLE II: Comparisons to the prior AI accelerators.

	A^3 [29]	SpAtten [30]	Sanger [31]	ViTCoD [32]	ReDAS [33]	This Work
Target AI Model	Transformer	Sparse Transformer	Sparse Transformer	Sparse ViT	General Purpose	ViT + CNN
Technology (nm)	40	40	55	28	28	28
Precision	INT9	INT12	4 & 16 bits	Not specified	INT8	INT8
Frequency (MHz)	1000	1000	500	500	1150	1850
Area (mm^2)	2.08	18.71	16.9	3.0	20.77	6.14
Power (mW)	110.422	2600*	2760	323.9	-	8330
Peak Performance (TOPS)	0.22	2.00	2.56	0.256	18.84	30.31
Compute Density (TOPS / mm^2**)**	0.11	0.11	0.15	0.09	0.90	4.93
Energy Efficiency (TOPS / W**)**	1.99	0.77	0.93	0.79	-	3.64

* Execluded the DRAM power for fair comparison.

TABLE III: Hardware Specifications.

Parameter	**Array Size**	**Total Area**	**Max Power**	**Max Frequency**	**Max Throughput**
Value	128×128	6.14 mm^2	8.33 W	1.85 GHz	30.31 TOPS

TABLE IV: Model configurations for hardware evaluation

Label	**ViT**	**CNN**	Param. Count (M)
A	TinyClip8	Resnet-18	19.96
B	TinyClip8	Faster R-CNN*	21.54
C	EfficientViT-m0	YOLOv4-tiny	24.41
D	TinyClip8	YOLOv4-tiny	24.13
E	EfficientViT-m5	Resnet-18	59.41
F	TinyClip32	Faster R-CNN*	74.67
G	EfficientViT-m5	YOLOv4-tiny	63.59
H	TinyClip32	YOLOv4-tiny	77.27

*Resnet-50 backbone (by fourth bottleneck block) used.

improvements. For instance, while both models improve over time, EfficientViT exhibits a sharper and more consistent quality enhancement, demonstrating its capability to bridge the gap between lightweight design and task-specific accuracy.

Subfigure (b) highlights the convergence behavior of TinyCLIP and the knowledge-distilled EfficientViT for Task 1. Knowledge distillation demonstrates not only faster convergence but also superior final performance compared to TinyCLIP. This result underscores the effectiveness of leveraging a teacher-student paradigm to enhance the compact model's capacity for complex visual understanding.

Overall, knowledge distillation enables lightweight models like EfficientViT to achieve task-specific accuracy comparable to larger models while maintaining efficiency. This approach ensures a robust and scalable solution for edge computing scenarios, as it balances computational constraints with the precision required for task-oriented object detection.

C. Energy Efficiency and Speedup Analysis

To evaluate compute performance, we configured the ViT/CNN pairs within the iTask framework as detailed in Table IV, with corresponding compute latencies presented in Fig. 7. The results demonstrate that our accelerator efficiently accommodates diverse configurations, spanning varying parameter counts and model architectures. This flexibility facilitates the selection of optimal configurations tailored to specific tasks and constraints, enhancing scalability and leveraging the strengths of both ViT and CNN configurations across a range of deployment scenarios.

Table II compares the state-of-the-art (SOTA) AI accelerators, showing our accelerator has better compute density and energy efficiency

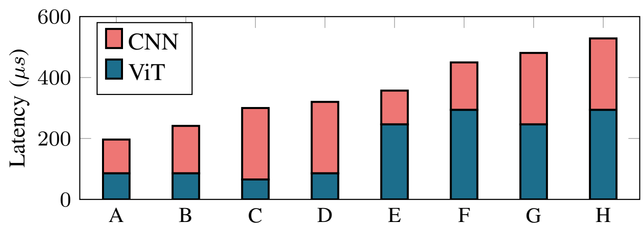

Fig. 7: The latency breakdowns on CNN/ViT architectures executed on the hardware accelerator.

than other works. This result originates from the dedicated design discipline for the iTask acceleration, including only the necessary modules and replacing on-chip SRAM with ping-pong SRs. At the same time, other accelerators adopted hardware sparse attention support to reduce the required throughput itself by pruning [30]–[32] or aware of general-purpose acceleration for most AI-related computations [33]. Another reason is that the newer semiconductor technology node, $28nm$, is used to synthesize our accelerator.

VI. Conclusions

In conclusion, iTask delivers a powerful framework for task-oriented object detection by integrating hardware acceleration, flexible model configurations, and situational awareness. Our hardware acceleration circuit significantly reduces latency and energy costs, making edge deployment feasible for real-time applications. The dual configurations—knowledge distillation for task-specific accuracy and quantization for versatile, multi-task generalization—offer adaptable performance that meets diverse task requirements without excessive computational demands. By enabling intelligent sensing, iTask selectively processes relevant data, optimizing resource use and providing an efficient, context-aware solution tailored for resource-constrained environments, from autonomous systems to real-time surveillance.

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